

$H_{\pi_0}^2(X)^{\mathbb{T}}$ belonging to the zero $(\mathbb{C}^\times)^n$ -weight space are represented by log-canonical bi-vector fields, we let

$$\mathcal{S}(\pi_0) = \text{the set of all the non-zero } (\mathbb{C}^\times)^n\text{-weights in } H_{\pi_0}^2(X)^{\mathbb{T}}, \quad (1.2)$$

and we focus on Poisson deformations of π_0 of the form $\pi = \pi_0 + \pi_1 + \pi'$, where π_1 is a sum of $\mathbb{T}$ -invariant algebraic bi-vector fields on X with $(\mathbb{C}^\times)^n$ -weights in $\mathcal{S}(\pi_0)$ and satisfies $[\pi_0, \pi_1] = 0$, and π' is a correction term such that π is Poisson. We show in §2.2 (see Lemma 2.8 and Lemma 2.9) that the set $\mathcal{S}(\pi_0)$ has the following properties:

1) $\mathcal{S}(\pi_0)$ is a finite subset of $(\mathbb{Z}_{\geq -1})^n$ and each $\theta \in \mathcal{S}(\pi_0)$ has exactly two entries of -1 ;

2) For every $\theta \in \mathcal{S}(\pi_0)$, the θ -weight space of $H_{\pi_0}^2(X)^{\mathbb{T}}$ is 1-dimensional with a basis vector represented by a uniquely defined bi-vector field V_θ on X (see (2.19)).

For $p \in [0, n]$, let $\mathfrak{X}^p(X)$ be the space of all algebraic p -vector fields on X , and let $[\cdot, \cdot]_{\text{Sch}}$ be the Schouten bracket on $\mathfrak{X}(X) = \bigoplus_{p=0}^n \mathfrak{X}^p(X)$. For $W \subset \mathbb{Z}^n$ and $p \in [0, n]$, let $\mathfrak{X}^p(X)^W$ be the direct sum of the $(\mathbb{C}^\times)^n$ -weight spaces in $\mathfrak{X}^p(X)$ with weights in W . By definition $\mathfrak{X}^p(X)^W = 0$ if $W = \emptyset$.

For a subset $\mathcal{S}$ of $\mathcal{S}(\pi_0)$ and $m \geq 1$, let $\mathcal{S}_{\geq m}$ denote the set of all elements in $\mathbb{Z}^n$ that can be written as the sum of m or more elements in $\mathcal{S}$. We say that $\mathcal{S}$ satisfies Condition (W1) if

$$\mathfrak{X}^1(X)^{\mathcal{S}_{\geq 2}} = 0.$$

Our main results in the first part of the paper are summarized as follows.

Theorem A. *Let π_0 be any $\mathbb{T}$ -log-symplectic log-canonical Poisson structure on $X = \mathbb{C}^n$, and let $\mathcal{S} \subset \mathcal{S}(\pi_0)$.*

1) (Theorem 2.21) *Suppose that $\mathcal{S}$ is $\mathbb{C}$ -linearly independent. Let $c = (c_\theta)_{\theta \in \mathcal{S}}$ be a set of formal commuting parameters, and let $\pi_1^{\mathcal{S}}(c) = \sum_{\theta \in \mathcal{S}} c_\theta V_\theta \in \mathfrak{X}^2(X)^{\mathcal{S}}[[c]]$. Then there exists $(\pi')^{\mathcal{S}}(c) \in \mathfrak{X}^2(X)^{\mathcal{S}_{\geq 2}}[[c]]$ such*

$$\pi^{\mathcal{S}}(c) = \pi_0 + \pi_1^{\mathcal{S}}(c) + (\pi')^{\mathcal{S}}(c) \in \mathfrak{X}^2(X)[[c]] \quad (1.3)$$

satisfies $[\pi^{\mathcal{S}}(c), \pi^{\mathcal{S}}(c)]_{\text{Sch}} = 0$. Moreover, such a $\pi^{\mathcal{S}}(c)$ is unique up to gauge equivalence (Definition 2.19);

2) (Theorem 2.26) *If $\mathcal{S}$ satisfies Condition (W1) (it is then $\mathbb{C}$ -linearly independent by Lemma 2.23), for any $c = (c_\theta \in \mathbb{C}^\times)_{\theta \in \mathcal{S}}$, there is a unique $\mathbb{T}$ -invariant algebraic Poisson structure $\pi^{\mathcal{S}}(c)$ on X of the form*

$$\pi^{\mathcal{S}}(c) = \pi_0 + \pi_1^{\mathcal{S}}(c) + (\pi')^{\mathcal{S}}(c) \in \mathfrak{X}^2(X), \quad (1.4)$$

where $\pi_1^{\mathcal{S}}(c) = \sum_{\theta \in \mathcal{S}} c_\theta V_\theta \in \mathfrak{X}^2(X)^{\mathcal{S}}$ and $(\pi')^{\mathcal{S}}(c) \in \mathfrak{X}^2(X)^{\mathcal{S}_{\geq 2}}$. Moreover, for any other $c' = (c'_\theta \in \mathbb{C}^\times)_{\theta \in \mathcal{S}}$, the two Poisson structures $\pi^{\mathcal{S}}(c)$ and $\pi^{\mathcal{S}}(c')$ coincide after a re-scaling of the coordinates $(x_1, \dots, x_n)$.

When $\mathcal{S}(\pi_0)$ is itself $\mathbb{C}$ -linearly independent as a subset of $\mathbb{C}^n$, the $\mathbb{T}$ -invariant formal power series Poisson deformation $\pi^{\mathcal{S}(\pi_0)}(c)$ of π_0 in (1.3) is said to be *maximal*, and we explain in Theorem 2.22 the precise sense in which it is maximal. Similarly, if $\mathcal{S}(\pi_0)$ satisfies Condition (W1), the $\mathbb{T}$ -invariant algebraic Poisson deformation $\pi^{\mathcal{S}(\pi_0)}(c)$ of π_0 in (1.4), for any $c = (c_\theta \in \mathbb{C}^\times)_{\theta \in \mathcal{S}(\pi_0)}$, is said to be *maximal*, and the precise sense in which it is maximal is explained in Theorem 2.34.

We remark that the miracle that the Poisson deformations along $\pi_1^{\mathcal{S}}(c)$ as in Theorem A are unobstructed has been observed in [MPS20] in the context of complex log-symplectic manifolds. While the methods of [MPS20] heavily used the homotopy algebra techniques, such L_∞ -algebras, our proofs are elementary and our strategies are adopted from [MPS24] where deformations of q -symmetric algebras are studied.

1.3 Applications to symmetric Poisson CGL extensions

In the second part of the paper, we apply Part 2) of Theorem A to a special class of examples.

Let again $\mathbb{T}$ be any complex algebraic torus with Lie algebra $\mathfrak{t}$ and character lattice $X^*(\mathbb{T}) \subset \mathfrak{t}^*$. We define an n -dimensional $\mathbb{T}$ -action datum to be a pair $(\langle \cdot, \cdot \rangle, \beta)$, where $\langle \cdot, \cdot \rangle$ is a symmetric $\mathbb{C}$ -bilinear form on $\mathfrak{t}^*$, and

$$\beta = (\beta_1, \beta_2, \dots, \beta_n)$$

is a sequence of n elements in $X^*(\mathbb{T})$ such that $\langle \beta_j, \beta_j \rangle \neq 0$ for every $j \in [1, n]$. Given such a pair $(\langle \cdot, \cdot \rangle, \beta)$, let π_0 be the log-canonical Poisson structure on $X = \mathbb{C}^n$ given by¹

$$\pi_0 = - \sum_{1 \leq j < k \leq n} \langle \beta_j, \beta_k \rangle x_j x_k \frac{\partial}{\partial x_j} \wedge \frac{\partial}{\partial x_k}, \quad (1.5)$$

¹While there is no harm to remove the minus sign in the definition of π_0 and adjust the results in the paper accordingly, the minus sign is there to align with the special example of standard Poisson structure on generalized Schubert cells which has appeared in a series of papers [LM17a, LM18, EL21] and has been implemented into a computer program.